

AI for energy optimisation

Applications in today's energy system

S U M M A R Y

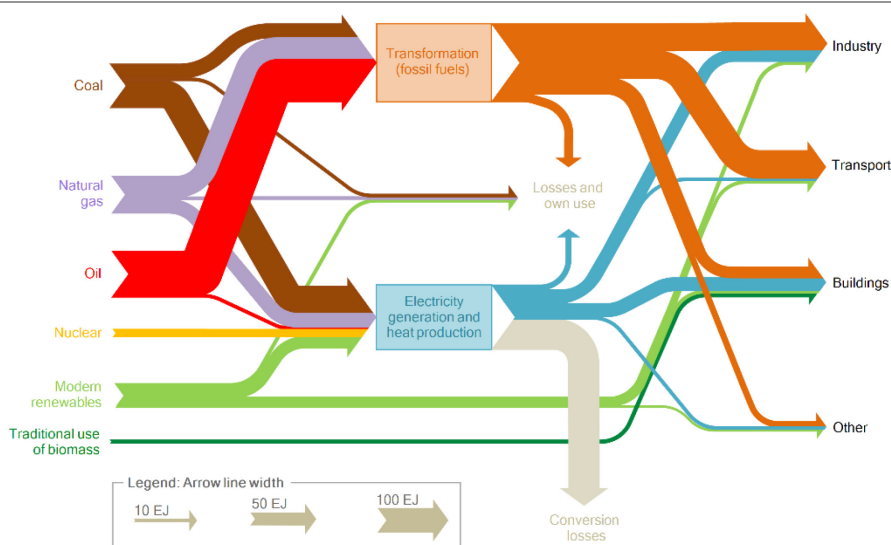
- The energy system is complex and evolving. It is becoming increasingly electrified, digitalised, connected and decentralised, with mounting cost pressures. These drivers have encouraged energy companies to deploy applications that utilise artificial intelligence (AI) to optimise systems, improve production, reduce costs, raise efficiency, cut emissions and enhance safety. In this chapter, we estimate the sector-wide impacts of *known* AI applications on a range of optimisations in a Widespread Adoption Case, to explore the impacts of optimistic uptake of AI in the energy sector.
- Oil and gas companies have been among the earliest adopters of new technologies to boost exploration and production. The number of supercomputers deployed in the sector has doubled since 2010 and total computing capacity has grown at almost 70% annually. In the Widespread Adoption Case, AI could reduce costs in oilfield development and operations, potentially improving the affordability of fuels, but it could also have broader ramifications, including increased emissions.
- AI could also have a major impact in electricity systems owing to the complexity of supply, transmission and demand profiles. In the Widespread Adoption Case, the application of AI in power plant operations and maintenance yields potential cost savings of up to USD 110 billion annually by 2035 from avoided fuels and lower costs. AI also enables greater integration of renewable electricity into the grid.
- The applications of AI in end-use sectors are varied but have significant potential. In industry, AI is being used to optimise production processes. In the Widespread Adoption Case, energy savings of around 8% could be achieved by 2035 in light industry, such as the manufacturing of electronics or machinery. AI in transport can enhance vehicle operation and management, which could cut energy consumption by up to 20%; it also has applications in reducing contrails and improving electric vehicle ranges. In buildings, the potential is limited by the rate of digitalisation, but there are compelling illustrations of impact, such as on efficiency and demand response.
- Accurate weather forecasts and analysis of changing weather patterns in a warming world are essential to optimise the operation, planning and resilience of energy systems. AI has been improving the accuracy of weather forecasts and also reducing computational demand.
- The adoption of such AI applications at a sector-wide level, however, is not a given. Various barriers are limiting the extent to which existing AI applications can be implemented, hindering the pace of change. These include unfavourable regulation, lack of access to data, inaccessibility, interoperability concerns, critical gaps in skills, the paucity of digital infrastructure and, in some cases, a general resistance to change.

3.1 Introduction

Artificial intelligence (AI) is being deployed across various parts of the global energy system, where AI applications are suited to meeting a wide variety of objectives, including cutting costs, integrating a growing share of variable renewables, making systems more efficient, enhancing fuel supply, ensuring timely maintenance of infrastructure and reducing emissions. This chapter focuses on where AI is being used, and could be used, to optimise or accelerate the deployment of *existing* technologies and processes used in the energy system. Chapter 4 examines the use of AI in the innovation process for *novel* technologies.

The energy system is highly complex, with multiple sources of energy following a web of flows and transformations to many end-uses. AI thrives on complexity like this, identifying patterns that can be leveraged to improve efficiencies. It is already having an impact on the energy sector but only in a limited, nascent way. Alongside greater electrification and digitalisation, AI is well placed to support a more resilient, affordable and sustainable energy future.

Figure 3.1 ▶ Energy supply, transformation and end-use, 2024



IEA. CC BY 4.0.

The global energy system is large and complex; fossil fuels still dominate primary energy supply, but their share is set to decline, while in end-uses the role of electricity is growing

The energy system can be understood in three broad parts:

- **Primary energy supply** includes the extraction or mining of energy resources.
- **Transformation and transmission** refers to the processing of primary energy sources into appropriate forms, such as the generation of electricity or the refining of crude oil, together with their transportation to consumers.

- **End-use consumption** represents the final consumption of energy for a desired outcome by consumers, such as to run a vehicle, heat something in a factory or cool a building. We categorise the end-use sectors as industry, transport, buildings and other.

Within each of these stages lie many applications, processes and techniques, each with its own set of challenges and opportunities for greater efficiency, security and sustainability.

In this chapter, we explore AI applications for energy resources (oil and gas, and mineral mining), electricity generation networks, and the end-use sectors of industry, transport and buildings. The chapter also discusses AI applications in weather forecasting and climate science and adopts a novel approach to understanding the broader sector-wide impacts of AI towards a wide range of optimisations. This approach is discussed in Box 3.1.

Box 3.1 ▶ Methodology adopted to assess AI's impact on the energy sector

Many of the desired goals of AI's application in the energy sector – such as cost reductions, enhanced reliability and improved resilience – are challenging to quantify at a broader sectoral level, beyond the confines of individual case studies. It is also challenging to predict the nature, adoption and impact of AI applications that might emerge in the future.

Given these limitations, this chapter introduces a new **Widespread Adoption Case**, which explores the impact that *known* AI applications could have at the sectoral level by 2035, assuming the widespread adoption of the application or technology. This case hinges on three considerations:

- The Widespread Adoption Case considers only existing AI-led interventions informed by real-world case studies that can be scaled to the sectoral level.
- It assumes that many of the existing barriers to the sector-wide adoption of these AI-led interventions (such as limited data availability and a lack of interoperability standards) are overcome.
- It stops short of considering the full theoretical potential of AI-led interventions, as it factors in certain insurmountable structural issues that would block their complete adoption. For example, we consider variations in adoption by region by factoring in the availability of enabling digital infrastructure.

Importantly, it is not a given that the Widespread Adoption Case will be achieved. Existing barriers, such as constraints on access to data and a lack of digital infrastructure and skills (discussed further in section 3.7), will continue to prevent widespread adoption in the absence of regulatory changes and incentives. Therefore, the Widespread Adoption Case is an ambitious pathway for the uptake of existing AI applications.

Note also that for the purpose of this analysis, we do not consider the impact of rebound effects. This issue is discussed further in Chapter 5, section 5.8. We also do not consider futuristic applications or interventions of AI in the sector, as their impacts are unknown.

3.2 The role of AI in the energy system

The energy sector is in a constant state of flux. The energy system is currently seeing rapid change that creates new challenges and opportunities – many of which are well suited to AI applications. The key trends in the energy sector include:

- **Rising electrification:** The overall share of total final energy consumption met by electricity has been steadily rising and is projected to accelerate.
- **Growing digitalisation:** Energy systems are becoming more digitalised and integrated through the proliferation of connected devices and appliances, electric vehicles (EVs), smart meters, and smart sensors in industrial and commercial applications.
- **Rising complexity:** The evolution of the energy system is resulting in greater complexity in supply, demand and energy flow patterns. On the supply side, electricity generation from variable sources, such as wind and solar, is growing fast. Generation is also becoming more distributed as smaller and more dispersed generation sources, such as rooftop solar, grow. On the consumption side, the number of connected appliances, vehicles and industrial facilities has been increasing. The result is a rise in the number of elements to manage both on the supply side and the demand side (Figure 3.2).
- **Pressure on costs:** The last few years have been challenging for energy consumers around the world, with high energy prices putting significant pressure on the cost of living. With new entrants in the market on both the supply and end-use sides, the energy sector has also become more competitive. These factors have been placing pressure on corporate finances, encouraging companies to find new ways to increase efficiencies and reduce costs.

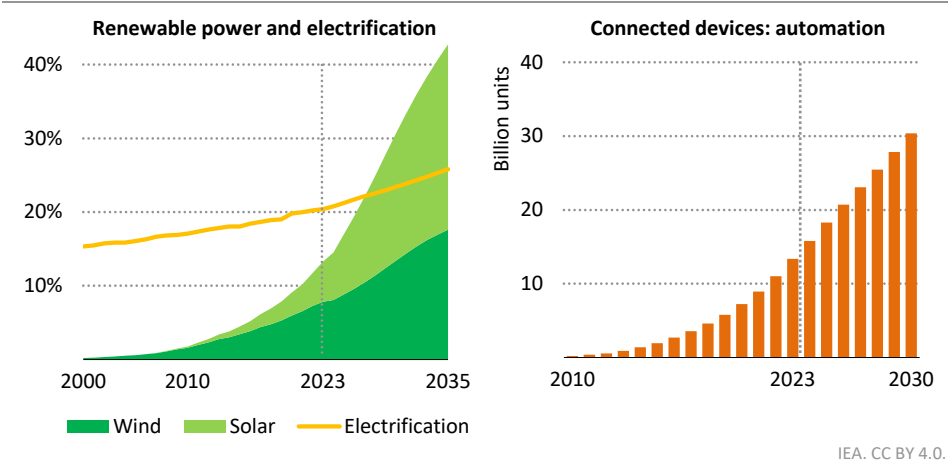
In addition to these structural trends, the energy sector is subject to several important policy objectives. International targets aim to make the energy sector more efficient and sustainable. The energy sector is the largest source of greenhouse gas emissions, which cause climate change. Energy sector emissions have continued to rise, reaching 37.8 gigatonnes of carbon dioxide (Gt CO₂) in 2024 – the hottest year on record (with 2023 the second hottest). Energy also needs to be reliable, affordable, secure and resilient. These imperatives have been cast into sharp focus by the energy market turmoil of recent years.

AI can help advance progress on these critical challenges, but its successful deployment is likely to depend on several key criteria. Typically, for AI applications to be deployed, they require the availability of digital infrastructure and skills. Widespread use of sensors, analytics and control systems allows for the collection of the extensive datasets that AI needs, with increased scope for automation. Where advanced software systems are already in place, AI capabilities can be rapidly deployed – but this is often inconsistent with the slow turnover of capital equipment in the energy sector.

The deployment of AI to solve energy challenges will also depend in part on the alignment of incentives. Uptake is likely to be strongest where the use of AI is in line with motives like

finding or harnessing more resources, reducing operational costs, cutting emissions, increasing resilience and boosting safety.

Figure 3.2 ▶ Shares of renewable power and electrification, and number of connected devices



The energy system is electrifying and becoming more complex as renewable sources grow; the number of connected devices is set to double from 2024 to 2030

Notes: Wind and solar show the share of global power generation; electrification shows the share of total final energy consumption that comprises electricity; pathway based on a scenario guided by today's policy settings. Connected devices shows the global stock of network-connected automation appliances.

Source: IEA 4E EDNA Total Energy Model V2.0 for Connected Devices (right graph).

The technological capabilities of AI continue to evolve. Currently, applications excel at learning from large and complex systems and applying their learnings to improve those systems (such as finding new resources) or enhance the control of them (such as daily operations). The system-wide, holistic aspect of these strengths suggests that AI applications will be most beneficial where implemented at scale, for example in interconnected power grids, large industrial facilities and commercial buildings. By contrast, for smaller, older assets with limited digital connections – such as many individual vehicles or residential homes – the role of AI may be more indirect or limited.

The range of potential applications of AI in the energy sector as surveyed in this chapter is broad (Table 3.1). At the simplest level, these applications can be summarised into two types: those that help to identify resources and design, plan and build facilities, and those that help to optimise, refine and automate the operation of energy systems. These can be applicable across the broad sweep of the energy sector – from identifying and harnessing resources, including fossil fuels and critical minerals for energy technologies, to the generation, transmission and distribution of electrical power and the use of energy in the buildings, industry and transport sectors. Indeed, many applications of AI have the potential to be

deployed across multiple sectors: for example, AI-enhanced digital twins or predictive maintenance can help optimise the design and operation of oil and gas platforms, large power plants and major industrial facilities. AI can also enhance the capabilities and accuracy of climate and weather science, yielding further potential energy sector benefits.

Table 3.1 ▶ AI applications for energy optimisation and their applicability by sector

Category	Oil and gas	Critical minerals	Power	Grids	Industry	Transport	Buildings
Resource management				n.a.		n.a.	n.a.
Applications related to the assessment, characterisation and extraction of resources, including fossil fuels, critical minerals, renewables (e.g. wind, solar, hydro and geothermal) and CCUS.							
Design and development							
Applications related to the design, planning, development and construction of assets to extract, harness, transform and transport resources, and assets that are end-users of energy.							
Operational optimisation							
Applications that enhance the efficiency and output of a process (or set of processes) related to the extraction, generation, transformation and transport of energy, or in end-use sectors.							
Automation and autonomy							n.a.
Applications that remove significant elements of human interaction within a system or process.							

Legend: n.a. = not applicable;  = limited relevance;  = moderate relevance;  = highly applicable
 Note: CCUS = carbon capture, utilisation and storage.

The rest of this chapter explores existing applications of AI across key parts of the energy system and their potential within each sector.

3.3 AI for energy and minerals supply

The extraction and supply of fossil fuels, nuclear fuel and the critical minerals needed for the components of energy equipment are the bedrock of the energy system. In this section, we explore the application of AI in optimising processes in the oil, gas and mineral extraction sectors.

Digitalisation in the oil and gas sector has progressed rapidly in recent years. Oil and gas companies were among the earliest adopters of supercomputers to boost prospects for oil and natural gas exploration and reduce costs. Mining companies have increasingly developed digital technologies in recent years. The growth of AI opens up the potential to expand on

this, helping companies to explore and identify additional volumes of oil, gas and minerals and plan their development, reduce costs, improve safety and reduce environmental impacts (Figure 3.3).

3.3.1 AI for oil and gas supply

The oil and gas industry has been a technology pioneer for more than 150 years and is now a complex global industry that can overcome large geological and engineering challenges. Continued investment in oil and gas supply will remain an essential element of energy transitions, even if demand were to decline in line with climate goals. This is because natural declines in oil and gas production from existing sources of supply are generally much sharper than declines in demand (a detailed discussion on this is provided in the IEA report, *The Oil and Gas Industry in Net Zero Transitions* (IEA, 2023)). AI applications in oil and gas supply can therefore help play a role in energy transitions by ensuring that sufficient supplies are available at lower cost and with lower emissions (Table 3.2).

Table 3.2 ▶ Applications of AI in the oil and gas sector

Application	Description	Impact on energy	Example
Resource management			
Exploration and development	More reliable resource evaluation; reduced predrilling uncertainty	● High: Reduced costs; faster development times	Subsurface data processing; reservoir simulation
Operational optimisation			
Operations and safety	Optimising and automating production and processes; leveraging digitalised set-ups	● High: Lower costs; greater reliability and resiliency through simplified supply chains; safer working conditions; fewer failures and environmental impacts	Remote operations; predictive maintenance; regulatory compliance
Emissions reduction	Better identify and mitigate leaks, both existing and at-risk	● High: More robust supply through improved leak detection, repair and prevention; long-term carbon storage certainty	Leak detection and repair automation and prediction; sensor data integration

In 2000, 11 supercomputers operated by oil and gas companies ranked among the world's 500 fastest. By 2024, this number had increased to 24, and total computing capacity has grown at almost 70% annually, outpacing the broader supercomputing industry. Companies including TotalEnergies, Petrobras and Saudi Aramco recently announced that they were developing new supercomputer capabilities for applications across exploration and

production, operations and safety, and emissions management; ENI's latest supercomputer is currently the fifth fastest in the world.

Oil and gas companies are also investing and partnering with AI experts to develop bespoke tools for their industry. For example, bp Ventures has made several investments in AI companies providing geological services since 2017, and ADNOC announced the completion of a 90-day trial of an AI agent based on a 70-billion-parameter large language model that it indicated improved the accuracy of seismic processing by 70%, along with other improvements (JPE, 2025). The Society of Petroleum Engineers, in collaboration with Aramco for financing, is delivering its catalogue of books and papers to a large language model, which will be commercially available in the near future.

Exploration and development

An essential part of exploring for and developing a new oil and gas deposit is characterising the subsurface by acquiring, processing and interpreting the results from seismic surveys. This is a data-intensive exercise – in the United Kingdom alone, the National Data Repository contains more than 130 terabytes (TB) of data from over 5 000 seismic surveys and other sources. The use of AI in seismic processing improves interpretation and image quality and makes it up to 90% better at classification (Araya-Polo, et al., 2017). After deciding to develop a project, companies need to decide where precisely to drill production wells, and this involves the collection of additional data from well logs and other images. The synthesis and interpretation of these datasets are increasingly being assisted by digital tools, such as machine learning, to help assess where the oil and gas may be present in sufficiently large accumulations.

Successful operations rely on simulating the behaviour of rocks and fluids during oil and gas production. Reservoir simulation models now use 2 TB to 10 TB of data and require systems capable of 100 teraflops to 1 000 teraflops of processing speed. AI can significantly enhance the accuracy and speed of these processes. The use of deep learning algorithms has allowed faster loading and processing of large volumes of data from multiple sources, including well logs, seismic data and production information, which are entered into simulation models. Physics-informed machine learning has enhanced the ability to model more complex reservoir behaviour (Anson, 2024). For example, Chevron combines field data with physics-based models and machine learning to predict well performance and production forecasts more accurately (JPT, 2022). This allows geological models of hydrocarbon reservoirs to be created in hours rather than months.

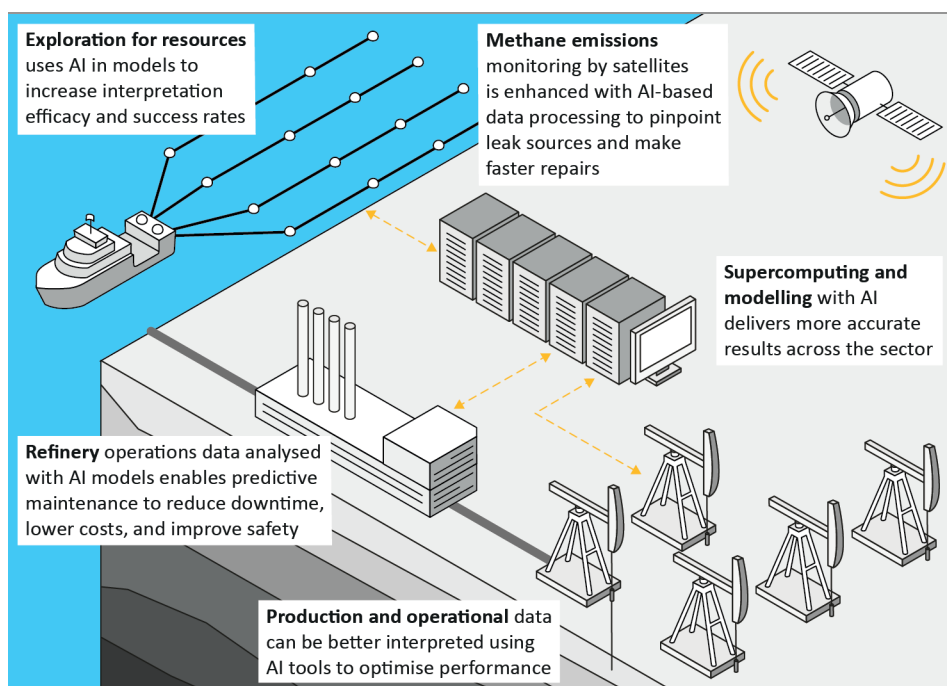
Operations and safety

Production forecasting is a critical component of the oil and gas industry, enabling companies to optimise operations and manage resources effectively. Traditional methods have been ever-present in the computational requirements of the industry, and they rely on many assumptions and oversimplifications. AI-driven forecasting methods have been evolving to overcome these challenges and improve results. Various AI and machine learning techniques

are being applied to production forecasting. For example, a hybrid AI model for oil production showed significant improvements in accuracy compared to traditional methods (Abdullayeva and Imamverdiyev, 2019), and a recent comparative analysis of machine learning techniques predicted oil production to a much higher degree of accuracy (Omotosho, 2024). Recently, ExxonMobil's AI-powered demand forecasting model was reported to have reduced forecast errors by 25% (Kuang, et al., 2021).

The use of AI can also significantly increase the potential for operations, monitoring and control to be carried out remotely. A typical oil platform operates tens of thousands of sensors (measuring aspects such as the temperature, pressure, and flow rates of produced liquids), which generate terabytes of data. Analysing and utilising these data streams from a centralised, remote location can increase efficiency and safety and reduce the costs of operations, which AI can assist in the management of (Figure 3.3). For example, cloud computing allows for the remote analysis of datasets, remote operational decisions and the creation of digital twins, such as Aker BP's recent streamlining of operations with digital twins.

Figure 3.3 ▶ AI applications in oilfield operations



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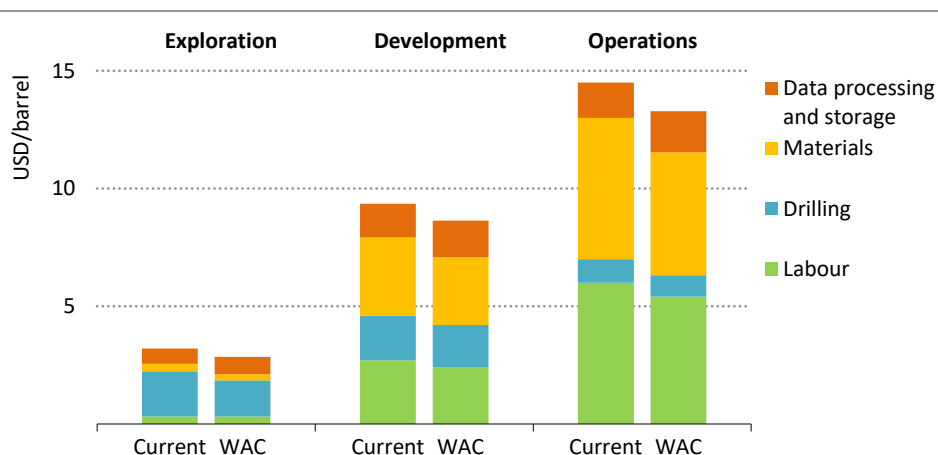
Many AI applications require input from sensors and the ability to process data remotely and quickly, supported by networks enabling data flows across geographies and systems

Assessing sector-wide impacts on costs

It is unlikely that AI can reduce the costs of all oil and gas production as things stand: many facilities around the world were installed some time ago and do not necessarily have the appropriate infrastructure to accommodate AI (retrofitting these facilities would carry additional costs, making the application of AI less attractive). Nonetheless, we can illustrate the potential of AI to reduce costs for new facilities by considering an example of new deepwater offshore oil development.

Producing oil from a new field involves labour, drilling, materials, and data processing and storage costs at each of the exploration, development and operations stages. A new offshore deepwater oil development today, with 25 million barrels of recoverable hydrocarbons, would cost around USD 10 per barrel in development and USD 15 per barrel during operations.

Figure 3.4 ► Cost of exploration, development and operations today and in the Widespread Adoption Case for a new oil deepwater project



IEA. CC BY 4.0.

In the Widespread Adoption Case, AI-led interventions could reduce the costs of finding, developing and operating a new deepwater offshore project by up to 10%

Note: WAC = Widespread Adoption Case.

We estimate that the widespread use of AI would mean drilling operations would become more efficient (e.g. fewer exploration wells would be required and production wells could be better targeted) (Figure 3.4). There would also be reduced labour needs (e.g. by allowing some operations to be carried out more remotely), and overall expenditure on materials would be lower (e.g. from more streamlined materials supply chains and less waste). Data processing and storage needs would increase substantially, but computing costs would become less expensive per unit of activity given economies of scale and the adoption of AI

processes. Overall, we estimate that in the Widespread Adoption Case (introduced in Box 3.1) deepwater costs could be reduced by up to 10%.

The reduction in the cost of oil production – in deepwater areas or elsewhere – does not necessarily imply a proportional reduction in the price of fuels at the pump, as that is determined by wider market forces as well as duties and taxes. Nonetheless, a marginal reduction in the oil price could lead to increased oil use, which in turn would have broader implications for greenhouse gas emissions. Section 5.8 in Chapter 5 includes discussion and analysis of these impacts.

Methane emissions reduction and carbon capture

The production and use of oil, gas and coal currently results in around 140 million tonnes (Mt) of methane emissions per year, or 4.2 gigatonnes of carbon dioxide equivalent (Gt CO₂-eq). This is around one-third of current anthropogenic methane emissions. A growing number of oil and gas companies have set methane targets, joining initiatives such as the Oil and Gas Methane Partnership 2.0 (OGMP 2.0), the Oil and Gas Climate Initiative (OGCI), and the Oil and Gas Decarbonization Charter (OGDC).

Despite commitments by countries and companies around the world, flaring and methane emissions from fossil fuel operations remain near record levels (IEA, 2024a). AI is now being deployed to boost data processing techniques to detect and quantify total emissions, both from major leaks over large areas and smaller leaks at the facility level. For example, automated AI-driven methane emitter monitoring systems using two satellites (Sentinel-2 and Landsat) were recently deployed at the International Methane Emissions Observatory's Methane Alert and Response System.

One particularly promising area is in rapidly detecting fugitive emissions, which comprise around 20% of methane emissions from oil and gas operations. These leaks can usually be repaired quickly once they are found, and the main challenge is finding them in an efficient and low-cost manner. Leak detection and repair programmes seek to do this, involving either equipping trained staff with optical gas imaging cameras or the use of airborne and satellite observations. AI can significantly improve the design and implementation of both of these approaches, including by reducing labour intensity and costs and improving the likelihood of finding leaks (Xia, Strayer and Ravikumar, 2024).

For the oil and gas industry, deep learning approaches enabled by AI allow data processing to classify emissions more quickly and predict future emissions to prevent leaks altogether (Bo, Zhang, and Liu, 2024; Aljameel et al., 2024; Wang et al., 2020). For airborne and satellite observations, AI allows the large amounts of data collected to be processed much more quickly to derive leak size and persistence. The use of remote sensing equipment also opens the possibility of continuously monitoring production facilities.

If the widespread adoption of AI could allow for continuous monitoring to take place at a far larger number of facilities and pipelines than is currently the case, this could help reduce emissions significantly, often at low cost. Based on data from the IEA's Global Methane

Tracker (IEA, 2024a), we estimate that if continuous leak detection and repair were to be implemented at sites that can currently only be examined quarterly or less frequently, this could avoid nearly 2 Mt of methane emissions globally (equivalent to around 60 Mt CO₂-eq). The costs of doing this would be around USD 1.6 per million British thermal units (MBtu), much lower than the average cost of USD 2.3 per MBtu of quarterly leak detection and repair programmes.¹

Another important possible deployment of AI is to improve the planning of carbon capture, utilisation and storage (CCUS) projects. Effective CCUS relies on subsurface knowledge and reservoir simulation. By enhancing reservoir models, additional computing power and AI can provide more certainty around the efficacy and costs of long-term CO₂ storage. Several oil and gas operators and service providers have been partnering with AI companies to improve carbon management, including Cerebras, a semiconductor and AI company that has partnered with TotalEnergies to improve carbon storage simulations (Cerebras, 2022).

3.3.2 AI for critical minerals supply

In the mineral mining sector, machine learning and AI techniques already play a significant role in exploration, mine operations and extractive metallurgy (Table 3.3). Many AI techniques in mineral exploration parallel those in upstream oil and gas industries, where machine learning has long been used for subsurface data interpretation, reservoir simulation and reducing uncertainty (Box 3.2). AI can be used to process geophysical data to improve anomaly detection and orebody prediction, lowering costs and boosting resource confidence while reducing sampling needs.

Once an ore deposit is identified, AI can contribute to improving productivity, safety and cost-efficiency in mining operations. Autonomous haulage systems allow for high-utilisation operations, reducing labour costs while increasing safety and fuel efficiency. Predictive maintenance algorithms analyse sensor data from heavy machinery to anticipate failures before they occur, helping to reduce unplanned downtime and extend equipment lifespans. AI is also being applied to ore tracking systems that monitor material movement from blasting through processing, ensuring that high-grade material is prioritised while minimising waste and environmental impacts.

Refining and metallurgical processes can also benefit from AI, which is driving gains in efficiency and recovery rates. Machine learning algorithms analyse real-time plant data, such as temperature, pressure and flow rates, to fine-tune processing conditions dynamically. Sensor-based sorting systems use AI to distinguish valuable ore from waste, improving pre-concentration and reducing the volume of material. Computer vision technology is being applied in flotation circuits to optimise mineral separation and recovery rates.

¹ These costs do not include the potential savings that accrue in many instances because the additional methane gas that is captured can often be sold or used.

Table 3.3 ► Key applications of AI across the mining life cycle

Application	Description	Impact on energy	Example
Resource management			
Exploration	Enhanced resource discovery, assessment and characterisation	● Low: Higher discovery success rates, lower costs, faster exploration timelines	Geophysical data analysis, remote sensing, geochemical modelling, drill target optimisation
Operational optimisation			
Mine operations	Enhanced automation and assessment of operations to improve efficiencies	● Low: Increased productivity and safety, reduced downtime and operational costs	Predictive maintenance, fleet dispatch, ore grade control
Processing and metallurgy	Enhanced use of data from real-time processing operations to gain efficiencies	● Medium: Higher recovery rates, lower energy and reagent consumption, improved process efficiency	Process automation, sensor-based sorting, machine vision in flotation, metallurgical modelling
Automation and autonomy			
Mine operations	Removal of human operation of haulage vehicles	● Medium: Increased productivity and safety, reduced downtime and operational costs, higher fuel efficiency	Autonomous haulage

Box 3.2 ► Reducing uncertainty in mineral exploration with AI

The Mingomba copper deposit in Zambia ranks among the largest undeveloped copper deposits in the world. It is estimated to contain about 250 million metric tonnes of copper at a grade of 3.6%, around seven times the grade of the average copper mine. (Lobito Corridor Investment Promotion Authority, 2024)

It was first discovered in the 1970s and planned as an extension to the Lubambe mine, located in the heart of the Zambian Copperbelt. Commercial copper mining in the region has been ongoing for more than a century, relying on its large, high-grade orebodies. Despite the resource potential indicated by the deposit’s proximity to existing reserves and commercial operations, the depth of the orebody – more than a kilometre underground – presented challenges in resource characterisation and recovery.

In 2022, KoBold Metals, a company specialising in AI-driven mineral exploration, acquired a stake in the Mingomba project and began applying its machine-learning and data-driven geoscience methodologies. In 2024, KoBold validated the conclusions of a 2020 concept study commissioned by Lubambe. Unlike conventional greenfield exploration, Mingomba provided a rich legacy dataset – including seismic surveys and historic drill core logs – which KoBold used to train its AI models.

Rather than relying on costly, high-density drilling, the AI model focused surveying efforts on areas that would yield the most useful data. This iterative process, comprising data

acquisition, model refinement and expert decision making, allowed geologists to improve resource estimates while minimising costs and environmental impact.

This illustrates the ongoing technological advances in mineral exploration in which AI and machine learning are augmenting geologists' decision making. As high-grade, near-surface deposits are depleted, expanding and rapidly assessing the search space for mineral resources is essential for secure, affordable mineral supplies.

AI models trained on multimodal datasets – including geophysical, hyperspectral and drilling data – can detect patterns imperceptible to traditional methods, improving the likelihood of success of both greenfield exploration and the reassessment of complex deposits. Most models are currently trained on specific geologies based on the available data, making the identification of significant resources elsewhere in the world beyond their capability.

3.4 AI for the electricity sector

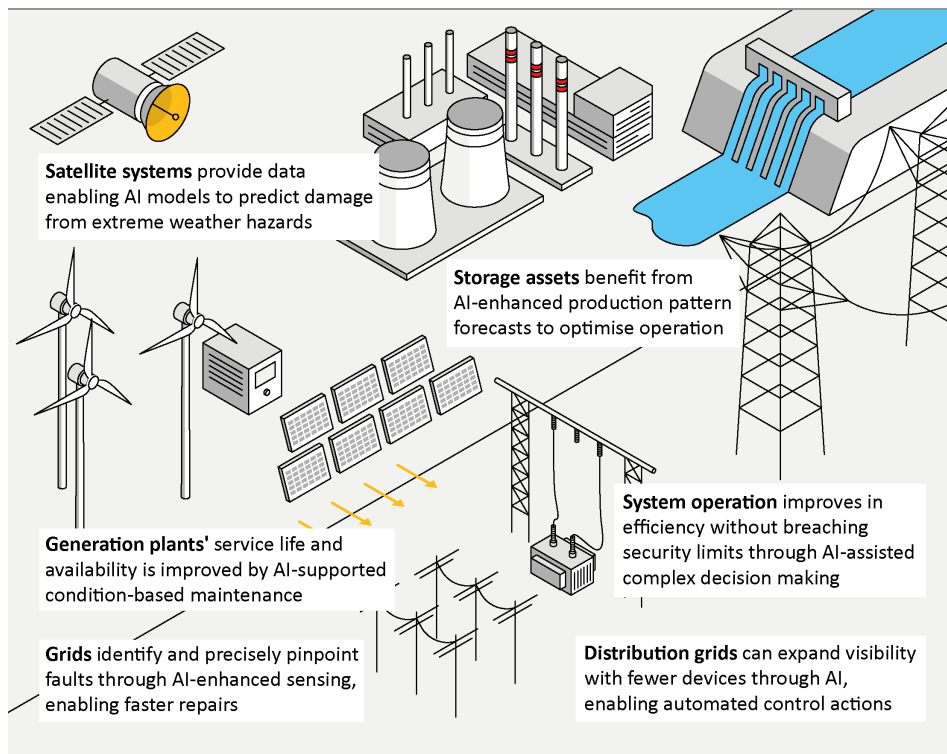
3.4.1 AI applications for power system operations

The electricity sector is on the brink of a significant transformation, facilitated by the rapid advancements in AI. Over the next decade, AI has the potential to play a pivotal role in the way power is generated, distributed and consumed, leading to increased efficiency, sustainability and resilience.

The power system has become increasingly complex in many countries, as the production of electrical power has shifted from large, centralised power plants to a multitude of small, distributed sources (Figure 3.5). In parallel, a digitalisation revolution is producing large pools of data, which in turn can be used to manage the complexity of the whole system. The integration of AI into the electricity sector could bring significant system-wide benefits with its ability to process huge amounts of data and provide optimisations based on trained models rather than predetermined rules. AI has the potential to play a critical role in managing the complexities of integrating renewable energy sources into the grid. AI-enhanced control systems could allow plants and facilities to operate at their rated performance for longer periods, improving efficiency while minimising downtime.

Managing the electricity system through traditional methods and rules – for example, direct communication with power plants to manage operations or the tiered response approach to frequency control – may still work in increasingly complex environments but would not take advantage of the potential of new technologies. As the number of sources of power system flexibility increases substantially, either in the form of energy storage or demand-side response, operational approaches need to be upgraded. Further examples are the fast-acting converters that are part of any solar photovoltaic (PV) array or wind turbine, and battery storage installations, which could be called on to a greater degree with more automated systems.

Figure 3.5 ▶ AI applications in electricity generation and transmission



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Power systems are increasingly complex, with more distributed sources and a wider set of flexibility sources, requiring more advanced operation methods that would benefit from AI

An important AI application at the system level is to enhance the forecasting of electricity demand and supply from variable renewables in order to optimise the use of power sector assets, including dispatchable power plants, energy storage and demand-side flexibility, and ultimately improve the overall efficiency of the power system. AI is already applied at the system level. For example, from the IEA Survey (see section 3.4.3) RTE in France and Elia in Belgium apply AI for real-time forecasting to assess system imbalances. The Nostradamus AI tool from Hitachi Energy provides accessible load, market price and renewable forecasts (Hitachi Energy, 2024). IBM Research established the GridFM working group to enhance power grid operations and planning with AI, focusing on resilience, efficiency and renewables integration using pre-trained optimal power flow models and multimodal data for outage prediction and load forecasting (IBM, 2024).

Advanced AI-driven weather and demand prediction models allow grid operators to anticipate fluctuations more accurately, minimising the curtailment of wind and solar PV in conjunction with demand shifting or storage. AI can significantly improve the accuracy of

weather forecasts by analysing vast amounts of historical and real-time meteorological data, which can also improve the resilience of energy systems (see section 3.6). For example, DeepMind's wind power forecast was found to increase the financial value of wind energy by as much as 20% (Google DeepMind, 2019).

Machine learning models can predict local weather conditions, such as wind speeds and solar radiation, with high precision. These accurate predictions help anticipate the output of wind and solar farms at specific locations. For example, in the United Kingdom, AI improved the National Grid ESO solar forecast significantly for up to 8 hours ahead (Fulton, et al., 2024), and KEPCO in Korea uses AI for wind speed prediction and to simulate real-time weather impacts on generation capabilities. Additionally, by optimising the integration of renewable energy sources, the reliance on fossil fuel-based power generation can be reduced. For example, reducing global average curtailment by a single percentage point in 2035 could cut demand by about 28 million tonnes of coal equivalent (Mtce) of coal and 14 billion cubic metres (bcm) of natural gas, avoiding approximately 120 Mt of CO₂ emissions.

The above applications of AI at the power system level can also enhance the efficiency of fossil fuel power plants. Thermal power plants, traditionally designed for continuous operation with high capacity factors, are experiencing a major shift in their operations. With increasing shares of wind and solar PV in many systems, the role of thermal power plants is evolving, with more emphasis on flexibility, which reduces their average efficiency. Such efficiency reductions can be minimised where the application of AI to power system operations allows improved scheduling so that flexibility-enabled thermal power plants can operate at higher utilisation rates. A single percentage point improvement in efficiency could reduce the fuel consumption and emissions of a coal-fired plant by 2.5% and a natural gas-fired plant by 2%. These measures not only optimise resource utilisation but also lower system costs and enhance grid stability.

3.4.2 AI applications for power plants and storage

The synergy between AI and digitalisation has the potential to lead to significant gains in the operation of power plants and storage. Potential applications of AI in power generation with significant impacts include faster research, process optimisation, dispatch prediction and service interventions (Table 3.4). Additionally, AI could enable better management of energy storage systems, ensuring that renewable power is effectively stored and released into the grid when needed.

AI-driven data analytics could improve planning, project design and real-time operational decisions, resulting in reduced fuel consumption, lower CO₂ emissions and extended asset lifetimes. Below are several examples of AI use cases in power plants and storage at different project stages.

Table 3.4 ► Potential applications of AI in power generation

Application	Description	Impact on energy	Example
Design and development			
Planning and design	Selecting the optimal equipment, siting and infrastructure planning for power plants and storage projects	● Medium: Maximise asset performance, lifetime operation and returns on investment	AI used to optimise the design of renewable projects such as wind and solar PV farms
Process optimisation	Processes necessary to deploy generation installations involve repetitive tasks	● Low: Productivity increases, generally resulting in cost savings	Generative AI analyses tender documents and supports the creation of proposals
Operational optimisation			
Dispatch prediction	Operators need to decide when to activate their assets	● Medium: Increases efficiency of the market, enables players to improve their business models	Market models representing the merit order system use AI to predict day-ahead market prices
Anomaly detection	Operational data need to be analysed both online and offline for unusual patterns	● Medium: Relieves human experts from routine tasks, increases productivity	AI is trained with normal signal patterns, as well as patterns from irregular events, and raises the alarm upon detecting irregularities
Service interventions	Service moving from predetermined inspection and maintenance schedules to more flexible ones	● High: Potential to reduce number of inspections during lifetime and/or extend lifetime of equipment	AI uses operational data from heat recovery steam generators to predict wear and corrosion status, thus reducing on-site inspection needs
Autonomous operation	Plants able to be maintained and operated with reduced staff	● High: Several intermediate stages exist to achieve fully autonomous operation; cost benefits increase with each step	AI uses sensor data, expert knowledge and even data collected by maintenance robots or drones to ensure safe and reliable remote or autonomous operation

Improving planning and design choices

At the earliest stage of a power plant or storage project, AI can be applied to make better choices concerning the planning and designs. For renewable energy projects, AI is being applied to design solar and wind projects, including the selection of primary equipment (solar panels or wind turbines), the siting of the equipment (orientation of panels, available areas and spacing of turbines) and the planning of supporting infrastructure, all to optimise performance and returns on investment. Examples of AI tools to optimise renewable project designs include the Wind Plant Graph Neural Network from the US National Renewable Energy Laboratory, the Sedar project by Iberdrola with the Barcelona Supercomputing Centre, Aurora Solar and the Google Maps Platform Solar API. For nuclear energy, AI is being

developed to improve reactor design and performance, benefiting from previous designs and experience, with the aim of reducing construction costs, extending operating lives, raising operational flexibility and enhancing safety. For battery storage, AI is being applied to optimise charging cycles and defect monitoring in order to extend asset lifetimes, reduce costs and maximise value.

The application of AI in the planning phase of thermal power plants relies on the digital data created during the operation of existing units. The typical number of sensors for a gas-fired power plant is more than 6 000. Depending on the sensor type, time resolution ranges from milliseconds to hours, and the amount of data points produced can be anywhere from a few thousand to several million per year for each sensor.

While rule-based algorithms can already extract much useful information from the pool of data produced by a power plant, using the data to train AI models adds many more potential use cases. Digital twins, which are virtual replicas of physical assets or processes, can benefit from AI's ability to make up for deficiencies in the representation, for example insufficient numbers of sensors, straying material properties or manufacturing tolerances.

Streamlining permitting and construction

In the next phase of a project, AI can also be applied to accelerate permitting and licensing processes, which often span several years and require thousands of pages of documents to be drafted by applicants and processed by regulators. Lengthy permitting times have been flagged as a concern for countries seeking to reach their policy ambitions, including for renewables. Recent efforts have been made to shorten timelines, including a rule in the European Union targeting a maximum limit of two years for wind power projects. AI tools can benefit from previous permitting processes and environmental data, enabling faster processes and better outcomes. For example, in the United States, a project is underway to train AI tools with information from over 28 000 documents related to close to 3 000 environmental impact statements (PNNL, 2024).

During the construction phase of power plants and storage, AI can be applied to streamline complex logistics and project planning, reducing construction times and costs. For example, wind turbine logistics are complex as a result of their size – a single blade can exceed 100 metres even for onshore projects. An AI machine learning tool from GE Vernova is targeting a 10% reduction in the logistical cost of installing wind turbines. Broader gains in technology supply chains would reduce construction times for a wide range of power plants and storage assets (GE Vernova, 2022).

Improving operational decisions

Once built, AI can be applied to improve the operation and maintenance of power plants and storage assets, offering the potential to increase uptime, improve efficiencies, reduce maintenance and fuel costs and even reduce stress on the asset. Predictive and condition-based maintenance strategies can potentially yield benefits in all these areas. These approaches rely on AI models trained with historic and asset-specific operational data. AI can

also facilitate the development of digital twins, such as those available from Siemens Energy, which have the potential to enable real-time monitoring and simulation of power sector assets, allowing for proactive maintenance and performance optimisation. AI could also be applied to key component inspections, with AI models trained to support visual screening by human experts.

AI can also be applied to optimise the operation of individual assets by using local weather and wholesale market predictions for demand and prices. Two examples of improved local weather forecasting are the MeteoFlow project by Iberdrola to optimise renewable energy output, applying big data, machine learning and AI (Iberdrola, 2016), and the Myst AI tool used by Enel to rapidly create highly accurate forecasts of renewable output (Enel, 2022). AI can also be used to predict hydropower inflow and generation, both in the short term (Sapitang, et al., 2020) and in the longer term, taking into account the effects of climate change (Salomon, et al., 2022). In the current energy landscape, thermal power plants are increasingly being utilised for load-following operations. This means that instead of operating at a constant output, these plants adjust their power generation to match the electricity demand. During periods of high renewable energy output, thermal plants reduce their output, and conversely, they ramp up production when renewable sources are insufficient to meet demand. This flexibility is crucial for maintaining grid stability and ensuring a reliable supply of electricity. Improved accuracy of predictions over minutes, hours, days and months could enable more optimal use of thermal and storage assets, enabling more continuous and efficient operations (Hanley and McGuire, 2023).

Expanding capabilities of power plants

Beyond the normal operation of power plants, AI has the potential to expand their accessible technical capabilities. Regulations governing the power sector and power plants have been developed over decades, including detailed codes and standards related to their technical capabilities. One example of this is the power factor requirement for an electric generator. Frequently, the generator rating (in mega-volt amperes) is required to be up to 25% above the driving turbine's capability (in megawatts). While there are scenarios that do require a certain margin, such as grid disturbances, most assets' mega-volt ampere utilisation is only a few percentage points above the turbine rating. AI tools could be applied to assess the real-time requirements in the grid surroundings and enable a reduction in the margin in many individual cases. This would allow smaller generators to be specified at lower cost.

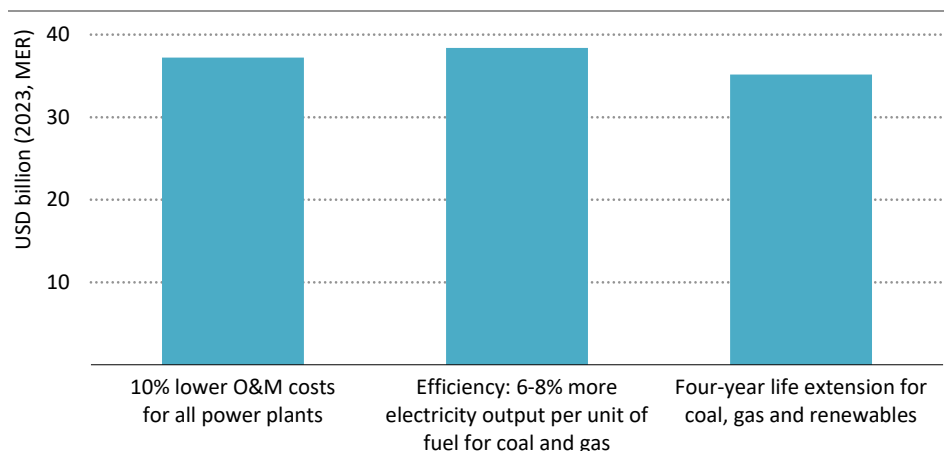
Power plants are also governed by customary requirements in tender documents that sometimes inhibit technical progress, like the application of AI. To unlock the extent of these opportunities, codes, standards and tender documents should be reviewed and adapted where appropriate.

Potential impacts of widespread adoption of AI in power generation

The widespread adoption of known AI applications in power plant operations and maintenance can yield substantial benefits. In the Widespread Adoption Case (see Box 3.1

for the methodology), AI has the potential to reduce operations and maintenance (O&M) costs by up to 10%, resulting in annual savings of approximately USD 40 billion (Figure 3.6). This is based on the assumption that 80% of costs are fixed O&M and 20% are variable O&M, with savings of 5% in fixed O&M through process automation and 10% in variable O&M through operation optimisation.

Figure 3.6 ▶ Illustrative potential annual cost savings in the Widespread Adoption Case in power plant operations worldwide, 2025-2035



IEA. CC BY 4.0.

The integration of today's AI applications in power plant operations and maintenance can yield potential cost savings of up to USD 110 billion annually worldwide to 2035

Note: MER = market exchange rate.

Efficiency improvements in fossil fuel power generation also play a crucial role. By optimising plant O&M schedules, average efficiency could increase by 3%, leading to annual savings of 200 Mtce of coal and 95 bcm of gas in the Widespread Adoption Case, while preventing an additional 850 Mt of CO₂ emissions. Enhancing efficiency to achieve 6-8% more electricity output per unit of fuel could be translated into fuel savings of around USD 40 billion per year, with 65% of the total savings coming from coal-fired power generation.

Extending the operational lifetime of power plants through AI could lead to considerable cost savings due to lower investment requirements. By prolonging the service life of all power plants by four years, the retirement of 435 gigawatts (GW) of capacity could be postponed by 2035, including 120 GW of wind and 50 GW of solar PV plants. This also includes the option of delaying the retirement of 170 GW of coal-fired plants and 60 GW of gas-fired plants, which could help avoid locking in new assets for decades. Consequently, up to 7% of cumulative investment in new power plants could be deferred during this period, amounting to USD 760 billion. This would save up to USD 35 billion annually in capital recovery payments (annual payments to recover the initial investment).

3.4.3 AI for electricity networks

Electricity grids are essential infrastructure that form the backbone of modern power systems, delivering electricity to homes, businesses and industries. These networks are evolving rapidly from traditional centralised systems into complex, digitalised networks that must safely accommodate variable renewable energy sources and distributed resources. Because these systems cannot tolerate failures – as interruptions can have widespread impacts on essential services – grid optimisation is increasingly important to improve efficiency and performance across operational parameters. The increasing complexity of grids demands significant investment in modernisation, expansion and digitalisation to prevent them from becoming bottlenecks to secure energy transitions. This includes deploying new transformers, power lines and pylons across both advanced economies and emerging market and developing economies to meet growing demand, replace ageing infrastructure and enhance resilience against extreme weather events and other disruptions.

3

Table 3.5 ► Potential applications of AI in the real-time operations of electricity networks

Application	Description	Impact on energy	Example
Operational optimisation			
Dynamic operating envelope	Framework that sets real-time, adjustable operating limits for grid-connected devices based on current network conditions to maximise available capacity while maintaining security; includes dynamic security assessment	● High: Reduces congestion costs, increases renewable integration, defers grid reinforcement investment and optimises existing infrastructure utilisation without breaching security limits	A grid operator increases line capacity by 15-30% during cooler weather conditions, safely accommodating additional renewable generation
Fault detection and localisation	Uses sensors and AI algorithms to quickly identify and pinpoint grid faults, reducing outage duration and improving response times	● High: Reduces outage duration by 30-50%, improves system reliability metrics (SAIDI/SAIFI), lowers restoration costs and enhances customer satisfaction	A distribution system operator detects a fault within seconds and precisely locates it within a 100-metre section, immediately dispatching repair crews to the exact location
State estimation and automation	Employs advanced algorithms to monitor distribution grid conditions in real time by inferring from measured points the electrical parameters at points without direct observability, enabling automated responses to maintain stability and optimise performance	● High: Improves grid stability during variable renewable generation, reduces operating margins, enables higher distributed renewables integration and decreases manual intervention requirements	An AI system continuously monitors voltage levels across the distribution network, automatically adjusting transformer tap settings to maintain optimal voltage profiles

Note: SAIDI = system average interruption duration index; SAIFI = system average interruption frequency index.

While AI delivers value across grid management, its most significant impact comes from short-term operational applications (Table 3.5). The increasing complexity of power systems demands advanced tools for two distinct challenges. First is ensuring human safety: real-time grid operations must prioritise human safety and reliable electricity supply above all else. Second is system optimisation: AI can help optimise the available capacity in the system – balancing generation, consumption and grid utilisation more efficiently in an increasingly variable environment. This approach delivers faster and potentially more cost-effective improvements without requiring new infrastructure investment. For long-term planning, AI helps navigate the substantial uncertainties in future electricity demand driven by widespread electrification, as well as the unpredictable evolution of power system technologies – from sophisticated grid solutions to emerging generation options – all while accounting for interactions with broader energy system developments.

Current AI adoption patterns in electricity networks

Despite higher potential benefits, short-term applications face greater resistance to AI adoption among grid operators. Our survey of grid operators from 13 countries, comprising Australia, Belgium, the People's Republic of China, Czechia, France, Germany, Ireland, Italy, Japan, Korea, Malaysia, the Netherlands and the United States, shows that only 23% use AI for real-time operations, while 54% have implemented AI for grid development planning and nearly 70% for asset maintenance and operation planning. AI applications in real-time operations focus on determining system imbalances, and load and generation forecasting, especially for renewables. Some operators expressed concerns about using AI in real-time operations, avoiding AI applications or limiting them to auxiliary assistants that advise operators in decision making. Operators use AI in asset operation planning to define and calibrate maintenance policies and activity planning, such as optimised scheduling and operation mode.

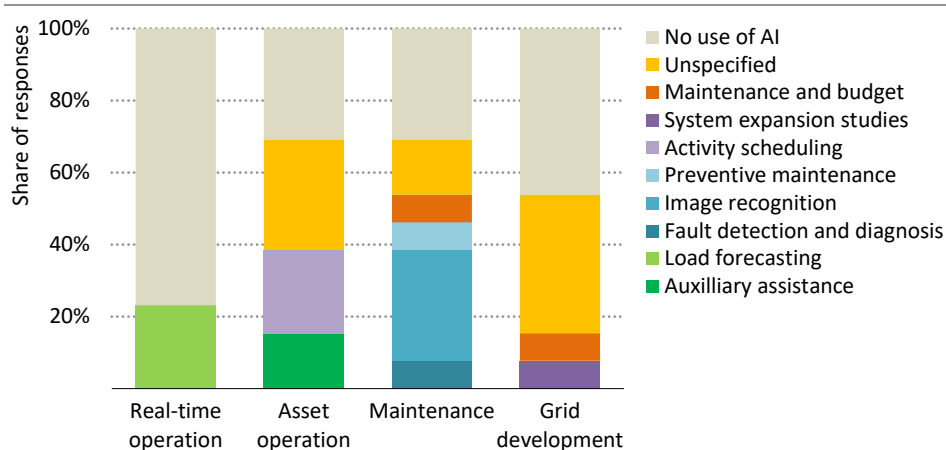
While real-time validation mechanisms for AI-driven grid decisions remain an emerging field with few established benchmarks or industry standards, there is greater receptivity to AI applications with extended decision time frames. Post-mortem analysis – the detailed investigation of power system failures and incidents to determine the root causes – allows for unhurried validation of AI-generated insights without operational time pressure. For example, 8% of respondents use generative AI for fault diagnosis.

Similarly, long-term scenario planning leverages AI's capacity to process datasets and identify patterns within complex energy system interactions across different time frames. Tapestry's Grid Planning Tool (Google X, 2024) enables large-scale, long-term grid simulations with high resolution, allowing grid planners to efficiently plan and reliably integrate renewable energy sources.

By using AI to run simulations, the power system operator in Chile can explore a wider range of scenarios and make informed decisions about grid investments and operations. AI effectively balances multiple competing objectives and criteria, a critical advantage in complex grid planning. From the survey, examples of Transmission System Operators (TSOs)

applying AI for planning include KEPCO in Korea, which uses AI for optimal energy planning and site selection and Elia in Belgium utilising graphics processing unit-based load flow computations for network planning.

Figure 3.7 ▸ Utilities using AI applications by category, 2024



IEA. CC BY 4.0.

AI's support for electricity grids today focuses on optimising asset operation planning and maintenance, including fault prevention by automated image recognition

In asset management, AI helps optimise maintenance schedules and equipment replacement based on condition monitoring, with decisions that can be methodically evaluated and refined. The survey results show that around 30% of respondents use image recognition AI to monitor and manage assets, including signal processing and the tracking of vegetation growth. For example, the State Grid Corporation of China (State Grid) applies reinforcement learning and AI to optimise scheduling and operation modes – advanced AI techniques that help manage complex trade-offs in grid asset operations. Among the TSOs, TEPCO in Japan, RTE in France, State Grid and the Elia Group all reported successful implementation of AI image analysis for asset monitoring and management, particularly for vegetation management. Hitachi Energy is one provider of such solutions (Hitachi Energy, n.d.). EirGrid is also exploring machine learning for predictive maintenance. Such applications could extend the operational lifespans of networks. For example, just two additional years in lifespan would defer the construction of over 3 000 km of new lines by 2035, reducing cumulative grid investment by USD 300 billion and saving around USD 15 billion annually in capital recovery payments.

However, adoption remains limited in system-critical operations despite their higher potential value. While some TSOs employ “auxiliary AI assistants” for decision support, real-time operations continue to rely on conventional tools and human expertise, with hesitancy

to implement AI solutions in time-sensitive scenarios, even where clear benefits have been demonstrated. As a result, some of AI's most valuable potential contributions remain underutilised in daily grid operations. Such caution is understandable for critical infrastructure. TSOs prioritise human oversight while gradually leveraging AI capabilities, maintaining essential reliability standards for the power system. Nevertheless, a balance is needed to preserve the necessary safeguards for critical infrastructure while accelerating adoption where appropriate to benefit from the efficiency gains and technological advancements that more comprehensive AI integration could deliver.

Implementation challenges and the way forward for AI in grids

The barriers to AI adoption in grid operations are common to those faced by all parts of the energy sector, as further discussed in section 3.7. Specific to grid operations, the barriers are predominantly institutional rather than technical, reflecting broader challenges in deploying new technologies rather than AI specifically. While some TSOs have embraced dynamic line rating (DLR) solutions, the majority of grid operators lag behind, despite DLR's proven ability to safely increase transmission capacity with clear technical benefits (see Box 3.3). This illustrates a potential pattern of resistance to innovation stemming from multiple factors: a cautious operational culture that favours maintaining current reliable practices over adopting new approaches, challenges faced in developing internal expertise that can bridge traditional power system knowledge and emerging technological capabilities, and a shortage of AI-skilled talent coupled with insufficient knowledge sharing.

Beyond institutional barriers, there are significant technical challenges with AI itself. In particular, AI systems often disregard physical laws and constraints, whereas electric utilities must adhere to these laws. The development of more mature and physically constrained AI solutions remains an area requiring improvement.

Regulatory frameworks further compound these challenges as they lack incentives for grid optimisation and struggle to evaluate novel technologies. Traditional regulatory models inadequately assess and manage risks from new approaches, while the transition from pilot projects to standard practice lacks structured support. Despite initiatives like regulatory sandboxes, TSOs receive insufficient backing for the R&D investments needed to establish new operational standards, creating a persistent innovation gap.

One of the significant limitations of relying solely on AI is that it falls short of providing transparent and auditable decision-making processes, as required by regulations. For example, event reports are mandated to document why human operators made specific decisions, but current AI systems lack the capability to articulate their own decision-making logic or provide a clear rationale for their actions in written documents.

Implementation requires thoughtful human integration. While AI excels at analysing datasets and recognising patterns, it cannot fully replicate human judgment in complex operational decisions. Historical data may not capture emerging system behaviours in evolving power systems. Explainability is key: engineers need to easily understand the underlying data and

assumptions to trust and act upon AI recommendations. Human expertise remains essential for understanding system-wide implications and managing unexpected situations, with ethical considerations and accountability necessitating clear human oversight. The challenge is to combine AI's analytical strengths with operators' contextual understanding to create robust, reliable systems.

Building on these implementation principles, accelerating AI adoption in grid operations could benefit from addressing several barriers simultaneously. Grid operators could consider creating dedicated teams with both power system and data science knowledge to address the data quality issues. EDP's Digital Factory (EDP, n.d.) has successfully integrated these skill sets to develop AI-based predictive maintenance and grid flexibility solutions. Knowledge sharing through peer networks and case studies offers ways to spread successful approaches across the sector. Examples include GO15 (GO15, n.d.), IEEE Power & Energy Society (IEEE, n.d.), Cigre (Cigre, 2022), the International System Operator Network (AEMO, n.d.), the Energy Systems Integration Group (ESIG, n.d.) and the Neural Information Processing Systems Foundation (Neural Information Processing Systems Foundation, n.d.), with which the French TSO RTE has been involved through its Learning to Run a Power Network competitions.

Beyond traditional funding, regulatory frameworks could evolve to include incentives rewarding improved grid capacity utilisation while maintaining reliability standards, as with the European Union Agency for the Cooperation of Energy Regulators (ACER, n.d.), Australian Energy Regulator (AER, n.d.) and the Federal Energy Regulatory Commission (United States, FERC, 2021). Well-designed regulatory testing environments, such as those in the United Kingdom (Ofgem, n.d.) and Singapore (EMA, 2024), present opportunities to bring new solutions to market more quickly. Training programmes combining power system fundamentals with AI literacy can help operators maintain appropriate control while benefiting from AI's analytical capabilities. These complementary approaches recognise that successful AI integration depends on both technical excellence and human expertise working in concert.

Box 3.3 ▶ Dynamic line rating in power grids: Unlocking unused capacity

Dynamic line rating (DLR) technology enables transmission lines to carry more electricity than their rated capacity. Instead of always using the same fixed limit, grid operators can adjust and safely expand the limit when weather conditions are conducive. For example, when it is cold or windy, the lines become physically cooler, allowing them to carry more electricity. Global experience shows that typically, transmission lines can safely carry 20-30% additional capacity above their maximum rating for around 90% of the time in any given year.

The value of AI with DLR lies in maximising the benefit of this additional capacity. When available, AI assists power system operators in deciding how to optimally use this capacity against other options, such as having to curtail renewables because of a lack of

grid transfer capacity, or building a whole new line in order to accommodate new peak flows. DLR technology does not necessarily rely on AI and has been tested and implemented by grid companies worldwide. However, real-time monitoring is crucial as actual conditions can occasionally fall below static ratings, creating undetected safety risks – making direct measurement vital for equipment safety.

This untapped resource could be mobilized quickly. DLR systems could activate 115-175 GW of additional global transmission capacity at a fraction of the \$35-52 billion cost of equivalent new power lines. DLR's core value comes from its ability to accommodate new power flows from increased demand or generation sources, effectively preventing bottlenecks that would otherwise require costly interventions. Nonetheless, barriers remain, including resistance from conservative utility cultures and the lack of regulatory incentives for efficiency innovations.

Not all lines require DLR, as bottlenecks are typically concentrated in specific sections. The French grid operator RTE estimates that equipping only 20 lines from its transmission fleet would maximise benefits. Grid operators should conduct cost-benefit analysis to identify priority candidates, considering deployment costs – typically a few hundred thousand dollars per line – against potential value. The economic calculation for determining how benefits are distributed among generators, consumers and operators varies by market structure and regulatory framework. For example, in the United States, on implementing DLR, PPL Electric Utilities (PPL, 2023) saved USD 65 million in one year by avoiding congestion costs on a single line (PV Magazine, 2025). In Belgium, socio-economic benefits of several thousands of euros have been reported in just hours, particularly when DLR enabled access to cheaper imports during periods of supply constraint (CURRENT, 2021).

For distribution circuits, the cost-benefit ratio has traditionally been less favourable due to lower electricity volumes. However, distribution grid companies like Arva in Norway managed to save EUR 30 million by avoiding a line upgrade to connect a wind farm by using a DLR system from Heimdall Power (Heimdall Power, 2022).

3.5 AI for energy end-uses

In addressing AI for end-users of energy, this report focuses on AI applications for energy optimisation in industry, transport and buildings. These sectors together account for around 95% of global end-use energy demand and have become increasingly digitalised and connected, unlocking the potential for AI-led optimisation.

3.5.1 AI for industry

The industry sector accounts for 39% of energy end-use and 45% of CO₂ emissions from energy. Energy-intensive industries account for more than two-thirds of industrial energy demand, using energy in processes to produce basic materials such as steel, cement, primary